

AI-Integrated Project Management System for Delay Cause Analysis Using Decision Tree

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Abstract. Effective project management plays a critical role in coordinating organizational resources, controlling project activities, and achieving strategic objectives. However, project delays remain a persistent challenge that can lead to budget overruns, inefficient resource utilization, reduced productivity, and diminished project quality. Although Artificial Intelligence (AI) and Machine Learning (ML) have increasingly been applied in project management, many existing studies primarily focus on project prediction, effort estimation, and risk assessment, while comparatively less attention has been given to delay cause analysis. In addition, the limited interpretability of many advanced AI models can reduce their practical value in managerial decision-making environments. To address these challenges, this study proposes an AI-Integrated Project Management System that combines project execution management with explainable AI-based delay cause analysis. The proposed architecture consists of an ASP.NET Core MVC web application, a SQL Server database, and an independent FastAPI-based AI service. Through the integration of project management, task management, progress tracking, resource management, cost management, and evaluation modules, the system provides a centralized environment for collecting and managing project data. These operational data are subsequently transformed into structured AI datasets for analytical purposes. The proposed framework employs a Decision Tree model to analyze project-related factors associated with delayed projects and identify potential delay causes. The model was selected because of its interpretability, suitability for tabular project data, and ability to provide understandable decision rules. To maintain transparency and managerial control, the system incorporates a Human-in-the-Loop validation mechanism in which AI-generated delay causes are reviewed and confirmed by project managers before being stored as validated knowledge. The main contributions of this study include the development of an integrated project management platform, the incorporation of an explainable Decision Tree-based delay cause analysis mechanism, and a continuous dataset enhancement process supported by manager-validated outcomes. The proposed framework demonstrates the feasibility of integrating explainable AI into operational project management workflows and provides a practical foundation for supporting delay cause analysis and project decision-making.

Keywords: Project Management System, Delay Cause Analysis, Decision Tree, Explainable AI, Decision Support.

1 Introduction

In the contemporary business landscape, effective project management serves as a fundamental mechanism for aligning organizational resources, managing constraints, and delivering strategic objectives. The ongoing process of digital transformation has increased the complexity of project environments, introducing multi-actor workflows, cross-functional collaboration, and increasingly sophisticated software development lifecycles. As a result, organizations face growing challenges related to task coordination, workflow visibility, and data management (Elri & Savaş, 2026; Li et al., 2022). To improve operational effectiveness, there is an increasing demand for data-driven project monitoring and decision support systems capable of providing a comprehensive view of project execution.

Despite the widespread adoption of digital management tools, project delays remain one of the most common challenges affecting project performance. Schedule deviations frequently contribute to cost overruns, inefficient resource utilization, and reduced deliverable quality (Sadeghi, 2024). In many organizations, project information is distributed across fragmented tools and disconnected information sources, creating data silos that limit visibility into the actual state of project execution (Elri & Savaş, 2026). Furthermore, delays often arise from multiple interacting factors, including inaccurate effort estimation (Elyas-sami & Idri, 2011; Nassif et al., 2015), personnel-related issues (Gupta & Suma, 2014), and procedural bottlenecks within project workflows. While modern analytical approaches have improved the ability to identify potential schedule risks, understanding the underlying causes of delays remains essential for implementing effective corrective actions.

To address these challenges, Artificial Intelligence (AI) and Machine Learning (ML) have increasingly been applied within project management. Previous studies have explored AI-based approaches for effort estimation, cost prediction, risk assessment, workflow monitoring, and decision support (Davahli, 2020). Advanced techniques such as Bayesian models, Long Short-Term Memory (LSTM) networks, and Graph Neural Networks (GNNs) have been investigated for analyzing project execution patterns and schedule-related risks (Kula et al., 2023; Mirjalili et al., 2025). In parallel, process mining approaches and AI-assisted workflow analysis have been employed to reconstruct execution processes and identify operational bottlenecks from historical project data (Dorado et al., 2025; Doropoulos et al., 2025; Cinkusz et al., 2025). As AI becomes more involved in managerial decision support, the importance of Explainable AI (XAI) and Human-in-the-Loop (HITL) approaches has received increasing attention. Transparent and interpretable analytical outputs are necessary to support practitioner trust and facilitate informed decision-making (Dam et al., 2018; Tantithamthavorn et al., 2020; Miller, 2023).

Although these developments have advanced the application of AI in project environments, several limitations remain. Many existing studies focus on predictive tasks, such as estimating project outcomes or forecasting schedule deviations, while comparatively less attention has been given to delay cause analysis. In addition, highly complex predictive models often provide limited interpretability, making it difficult for project managers to understand the reasoning behind a recommendation. Existing root

cause analysis approaches may also struggle to distinguish causal relationships from statistical associations, potentially reducing the reliability of analytical conclusions (Rayegan & Menzies, 2026). Furthermore, root cause analysis is frequently conducted as a separate analytical activity rather than being integrated directly into operational project management systems (Jin et al., 2024).

To address this gap, this research proposes the development of an AI-Integrated Project Management System that supports delay cause analysis through an interpretable Decision Tree model. Decision Trees were selected because of their transparency, suitability for tabular project data, and ability to provide understandable decision rules that support managerial interpretation (Dhebar & Deb, 2020; Mantovani et al., 2023). The proposed system combines project execution management with AI-assisted analysis, enabling project data to be transformed into structured information for identifying potential delay causes.

The main contributions of this study are threefold. First, it presents an integrated project management platform that centralizes project, task, resource, progress, and cost information. Second, it incorporates an independent AI service dedicated to delay cause analysis. Third, it adopts an explainable Decision Tree model combined with a Human-in-the-Loop validation mechanism, ensuring that AI-generated delay causes are reviewed and confirmed by project managers before being incorporated into future training datasets.

The remainder of this paper is organized as follows. Section 2 discusses Project Management and Delay Management. Section 3 presents the design of the proposed AI-Integrated Project Management System. Section 4 describes the Decision Tree-based delay cause analysis approach. Section 5 discusses the experimental results and system evaluation. Finally, Section 6 concludes the paper and outlines future research directions.

2 Project Management and Delay Management

Effective project management serves as the structural backbone for modern organizations, enabling them to align resources, manage constraints, and deliver strategic value. However, managing complex workflows, particularly in environments involving cross-organizational collaboration or software development, presents significant challenges (Elri & Savaş, 2026; Li et al., 2022). One of the most persistent issues in this domain is project delay, which frequently leads to cascading negative impacts, including severe cost overruns, compromised quality, and misaligned resource allocation (Sadeghi, 2024). Delays rarely stem from a single isolated event; rather, they are the culmination of interrelated factors such as inaccurate effort estimation (Elyassami & Idri, 2011; Nassif et al., 2015), suboptimal personnel selection and team volatility (Gupta & Suma, 2014), continuous scope changes, and procedural bottlenecks during approval workflows.

The traditional paradigm of project management often relies on fragmented tools—such as isolated spreadsheets, manual tracking mechanisms, and unstructured communication channels—which create data silos and obscure the true health of a project. This

fragmentation prevents project managers from gaining a holistic view of the project lifecycle, making it exceedingly difficult to perform accurate root cause analysis (RCA) when delays occur. In recent years, the integration of Artificial Intelligence (AI) and Machine Learning (ML) into project management has emerged as a critical evolution to address these systemic inefficiencies. The current state of AI in project management encompasses various applications, ranging from effort prediction and cost estimation to risk analysis and workflow optimization (Davahli, 2020). Advanced methodologies, such as the deployment of Large Language Models (LLMs) and multi-agent systems, are being explored to translate unstructured team dialogues into structured project artifacts, thereby reducing administrative overhead and capturing critical operational data (Doropoulos et al., 2025; Cinkusz et al., 2025).

A core requirement for effective delay management is the ability to leverage historical and real-time project data to support monitoring and decision-making. Through continuous data collection, organizations can construct comprehensive event logs that document every state transition, resource reassignment, and financial adjustment. These logs form the foundation for Process Mining, a technique used to reconstruct actual workflow executions, identify bottlenecks, and quantify deviations from planned schedules (Dorado et al., 2025). By applying AI to these structured logs, predictive process monitoring can provide early warnings of potential deadline compliance failures. For instance, dynamic prediction models that incorporate Bayesian modeling and delay pattern analysis have proven highly effective in adapting to execution changes and forecasting delays in software projects (Kula et al., 2023).

However, while predicting that a project will be delayed is valuable, understanding why it is delayed is crucial for implementing corrective actions. This necessitates a shift from purely predictive analytics to Root Cause Analysis (RCA). Research emphasizes the need to break the cycle of recurring failures by applying generative AI and structured analytics to uncover systemic root causes rather than merely treating superficial symptoms (Jin et al., 2024). A major challenge in RCA is distinguishing between correlation and causation within complex software analytics models. Correlation-based models can sometimes conflate associated variables with true causal drivers, leading to unstable or misleading explanations (Rayegan & Menzies, 2026). Therefore, the adoption of AI in project management must be accompanied by robust Explainable AI (XAI) frameworks. Explainability ensures that AI models provide transparent, interpretable, and actionable insights, which is a fundamental requirement for fostering trust among human practitioners (Dam et al., 2018; Tantithamthavorn et al., 2020). Ultimately, the goal is not to replace human judgment with autonomous black-box algorithms but to facilitate "Evaluative AI"—a hypothesis-driven decision support paradigm where AI presents structured evidence for or against specific delay causes, allowing human managers to calibrate their trust and make informed, final determinations (Miller, 2023).

To contextualize the proposed approach within the broader academic landscape, Table 1 provides a comparison of recent studies addressing delay analysis, machine learning applications, and explainability in project environments.

Table 1. Comparison of Related Studies.

Author(s)	Domain	Method	Explainable AI Focus	Delay/Root Cause Analysis
Davahli (2020)	<i>Project Management</i>	<i>Machine Learning (Various)</i>	<i>No</i>	Yes (Prediction)
Kula et al. (2023)	<i>Agile Software Projects</i>	<i>Bayesian Regression, Clustering</i>	<i>Yes</i>	Yes
Dorado et al. (2025)	<i>Software Workflows</i>	<i>Process Mining, LSTM</i>	<i>No</i>	Yes
Tantithamthavorn et al. (2020)	<i>Software Engineering</i>	<i>Random Forest, LIME</i>	<i>Yes</i>	Yes
Jin et al. (2024)	<i>Legacy Banking IT</i>	<i>Generative AI, Knowledge Graphs</i>	<i>Yes</i>	Yes
Rayegan & Menzies (2026)	<i>Software Analytics</i>	<i>Causality-aware Decision Trees</i>	<i>Yes</i>	Yes

Although existing project management platforms provide comprehensive support for project planning, task tracking, and team collaboration, AI-assisted delay cause analysis remains largely absent from mainstream project management environments. To further illustrate this gap, Table 2 compares representative project management platforms with the proposed system.

Table 2. Functional Comparison of Project Management Platforms.

Feature	Trello	Jira	Odoo	Proposed System
Project Management	✓	✓	✓	✓
Cost Management	✗	Partial	✓	✓
Delay Cause Analysis	✗	✗	✗	✓
Explainable AI	✗	✗	✗	✓

The comparison demonstrates that while existing platforms offer strong support for project execution and collaboration, they rarely incorporate explainable AI capabilities for identifying the causes of project delays. This limitation motivates the development of the proposed AI-Integrated Project Management System, which combines project management functionality with transparent AI-assisted delay cause analysis.

3 Design of the AI-Integrated Project Management System

To operationalize data-driven RCA, the proposed solution is an enterprise-grade AI-Integrated Project Management System designed to govern end-to-end project execution while providing intelligent decision support for analyzing de-lay causes. The system is engineered around a strict separation-of-concerns principle, ensuring that AI integration does not disrupt core operational workflows.

The overall architecture adopts a modern three-tier client-server model. The presentation and business logic layers are built upon the ASP.NET Core MVC framework,

which acts as the central orchestration engine. This layer handles user authentication, claim-based authorization, state transitions, data validation, and the aggregation of operational metrics. The data layer utilizes Microsoft SQL Server to persist all operational records, including hierarchical project structures, financial ledgers, and comprehensive audit logs. The AI component is decoupled into an independent compute microservice developed using Python and FastAPI. This AI service communicates with the MVC application via strict REST API contracts, focusing exclusively on dataset validation, model training, and delay-cause inference without possessing direct write access to the operational data-bases.

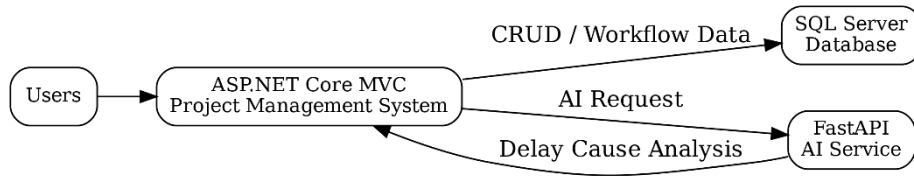


Fig. 1. Overall System Architecture (This figure illustrates the 3-tier architecture, showing the user interaction with the ASP.NET Core MVC web application, the data persistence flow to the SQL Server Database, and the REST API integration with the Python FastAPI microservice dedicated to AI compute tasks.).

The system is divided into several interconnected functional modules that collectively capture the operational realities of project management:

1. **Project Management:** Governs the end-to-end lifecycle of projects, strictly enforcing state transitions from initialization through execution to final completion.
2. **Task Management:** Manages granular work items. Task statuses are automatically synchronized upwards, ensuring that the completion of individual task details accurately reflects on the parent task and the overall project state.
3. **Progress Tracking:** Facilitates the submission and review of progress reports. Updates are governed by a rigid approval workflow, ensuring that only verified reports impact the effective completion metrics.
4. **Resource Management:** Controls team formulations and individual personnel assignments, meticulously logging workforce volatility and leadership transitions over time.
5. **Cost Management:** Manages financial constraints through budget modification proposals and actual cost tracking against approved financial baselines.
6. **Project and Employee Evaluation:** Supports post-execution appraisals based on predefined criteria, linking individual operational performance directly to project outcomes.
7. **AI Delay Cause Analysis:** An analytical module that extracts knowledge from historical executions to hypothesize primary causes of delay, providing actionable intelligence to managers.

The integrity of the AI model is highly dependent on the quality of the data collection process. Every critical action within the operational modules—such as task reassignments, budget modifications, or rejected progress reports—is immutably recorded in

system event logs. These logs provide a rich temporal dataset that reflects the true friction and bottlenecks experienced during execution, serving as a reliable foundation for workflow analysis and feature extraction.

To construct the required datasets, the MVC application executes an aggregation pipeline that synthesizes data from the operational tables into a flat, tabular feature vector for each project. A crucial data-governance rule enforces that only projects objectively classified as "delayed" and possessing a manager-confirmed delay cause are utilized for model training. The system explicitly prevents the use of raw, unverified AI predictions as ground-truth labels. Instead, it relies on a dedicated structure that stores the final, human-validated reason. This mechanism ensures that the training labels represent verified business realities.

The interaction between the project management system and the AI service is strictly designed to empower the human manager. When an analysis is initiated for a delayed project, the operational system packages the project's feature vector and transmits it to the AI endpoint. The AI service processes this vector through its trained model and returns a predicted delay cause along with a confidence score. If the confidence falls below a predefined threshold, the system triggers a fallback rule mechanism to prevent unreliable outputs. The manager then reviews this AI-generated hypothesis alongside the system's operational dashboards. The AI acts purely as a decision support tool; the manager retains full authority to accept the AI's suggestion or override it based on unquantifiable contextual knowledge. Once confirmed, the decision is logged, continuously expanding the high-quality dataset for future model retraining.

4 Delay Cause Analysis Using Decision Tree

Within the domain of AI-assisted project management, the selection of an appropriate machine learning algorithm must carefully balance predictive accuracy with human interpretability. While neural network architectures can identify highly complex patterns, their "black-box" nature obscures the reasoning behind their predictions, rendering them unsuitable for RCA where stakeholders require transparent justifications. Consequently, this system utilizes the Decision Tree algorithm for delay cause analysis. Decision Trees are fundamentally "white-box" models that partition the feature space using explicit, hierarchical IF-THEN rules, directly mirroring the logical deduction processes employed by human managers (Dhebar & Deb, 2020).

The construction of the Decision Tree relies on metrics such as Entropy and Information Gain to determine the optimal splitting criteria at each node. Entropy measures the level of impurity or uncertainty within a dataset S , defined mathematically as:

$$Entropy(S) = \sum_{i=1}^c -p_i \log_2(p_i) \quad (1)$$

where p_i is the proportion of samples belonging to class i . To evaluate the effectiveness of a specific attribute in classifying the data, Information Gain is calculated as the expected reduction in entropy:

$$IG(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) \quad (2)$$

By recursively maximizing Information Gain, the algorithm constructs a tree that isolates the most discriminative factors contributing to project delays. Entropy measures the uncertainty of a dataset, while Information Gain quantifies the reduction of uncertainty obtained by splitting the data using a specific feature. During model training, the attribute with the highest Information Gain is selected as the splitting criterion, allowing the Decision Tree to identify the most influential factors associated with project delays.

The choice of Decision Trees is heavily supported by literature in software analytics and process monitoring. Research demonstrates that rule-based models and classification trees offer a high degree of transparency, allowing practitioners to easily trace the causal chain of a prediction (Dam et al., 2018). Furthermore, Decision Trees are inherently well-suited for tabular data—the primary format of aggregated project metrics—and can effectively handle non-linear relationships without requiring complex feature transformations (Oca et al., 2020; Mantovani et al., 2023). While other approaches such as Graph Neural Networks (GNNs), Long Short-Term Memory (LSTM) sequence models, or complex ensemble kernels can capture sophisticated temporal and topological dependencies (Mirjalili et al., 2025; Kula et al., 2023; Feng & Baumgartner, 2020), they often sacrifice the immediate interpretability essential for administrative trust. Advanced tree variants, such as statistically-driven oblique trees (Li et al., 2025) or Temporal Logic trees (Liang et al., 2024), demonstrate the algorithm's vast potential, but the standard Decision Tree framework remains optimal for establishing a reliable, baseline explainable system (Pei et al., 2025; Keck, 2016).

The input features for the Decision Tree model are engineered to capture a holistic view of the project's schedule, resources, budget, and administrative friction. The MVC application aggregates 22 distinct features from the operational database, as detailed in Table 4.

Table 4. Input Features for the Delay Cause Analysis Model.

Feature Variable	Source Domain	Description
SoNhanVienDuAn	Resource	Total number of distinct personnel assigned to the project.
TongSoCongViec	Task	Total number of tasks defined within the project scope.
SoCongViecTre	Task	Absolute count of tasks that missed their planned deadlines.
TyLeCongViecTre	Task	Ratio of delayed tasks to total tasks.
ChiPhiDuKien	Financial	The total planned budget based on approved financial versions.
ChiPhiThucTe	Financial	The aggregate of actual costs incurred during execution.
ChenhLechChiPhi	Financial	The variance between actual costs and the planned budget.
SoLanThayDoiNhanSu	Resource	Frequency of personnel additions, removals, or reassignments.
SoLanThayDoiQuanLy	Resource	Count of approved project manager transitions.
SoNgayTreTienDo	Schedule	The cumulative deviation in days from the baseline project schedule.
SoDeXuatCongViecChoDuyet	Workflow	Volume of task proposals currently pending managerial approval.
SoDeXuatCongViecBiTuChoi	Workflow	Volume of task proposals rejected by management.
ThoiGianDuyetCongViecTrungBinh	Workflow	Average processing time (in days) required to approve task proposals.
SoDeXuatNganSachChoDuyet	Workflow	Volume of budget modification proposals pending approval.
SoDeXuatNganSachBiTuChoi	Workflow	Volume of budget modification proposals rejected.
ThoiGianDuyetNganSachTrungBinh	Workflow	Average processing time (in days) required to approve budget changes.
SoBaoCaoTienDoChoDuyet	Progress	Count of progress update submissions awaiting verification.
SoBaoCaoTienDoBiTuChoi	Progress	Count of progress update submissions rejected due to inadequacy.
SoBaoCaoTienDoYeuCauBoSung	Progress	Count of progress updates returned to employees for additional details.
TyLeBaoCaoTienDoBiTuChoi	Progress	The ratio of rejected progress reports to total submitted reports.
SoLanCapNhatTienDo	Progress	The total frequency of progress reporting activities.
SoNgayChamCapNhatTienDo	Progress	The latency (in days) since the most recent progress update was recorded.

Table 5. Top Features Influencing Delay Cause Analysis.

Feature	Importance
Planned Cost (ChiPhiDuKien)	0.2041
Delayed Progress Update Days (SoNgayChamCapNhatTienDo)	0.1562
Schedule Delay Days (SoNgayTreTienDo)	0.1460
Cost Variance (ChenhLechChiPhi)	0.1279
Number of Manager Changes (SoLanThayDoiQuanLy)	0.1236
Delayed Task Ratio (TyLeCongViecTre)	0.0619
Number of Staff Changes (SoLanThayDoiNhanSu)	0.0502
Actual Cost (ChiPhiThucTe)	0.0407

The feature-importance analysis shows that budget-related, schedule-related, and managerial factors contribute most significantly to delay cause identification. According to the active model metadata, planned cost, delayed progress update days, and schedule deviation days were identified as the most influential features contributing to delay cause classification. These findings are consistent with previous studies indicating that project delays are often associated with cost pressure, ineffective progress monitoring, and management instability.

The target variable for classification is derived from a standardized taxonomy of 10 delay causes, representing common points of failure in organizational workflows, as detailed in Table 6.

Table 6. Delay Cause Categories.

Category Code	Delay Cause Description
Lack of Personnel	Delay associated with insufficient or unstable staffing capacity.
Continuous Requirement Changes	Delay pressure caused by changing project requirements or scope creep.
Approval/Process Delay	Delay linked to internal approval bottlenecks or procedural friction.
Budget Overrun	Delay context associated with budget over-consumption or financial constraints.
Technical Risk	Delay due to technical uncertainty, integration issues, or implementation risk.
Dependency/Coordination Delay	Delay propagated from weak inter-task coordination or blocked dependencies.
Insufficient Input Information	Delay due to incomplete input artifacts, missing prerequisite data, or ambiguous briefs.
Inaccurate Time Estimation	Delay resulting from planning or effort estimation errors during initialization.
Incomplete Progress Updating	Delay linked to weak progress update discipline and communication gaps.
Other	Residual category utilized when no dominant predefined causal pattern is confirmed.

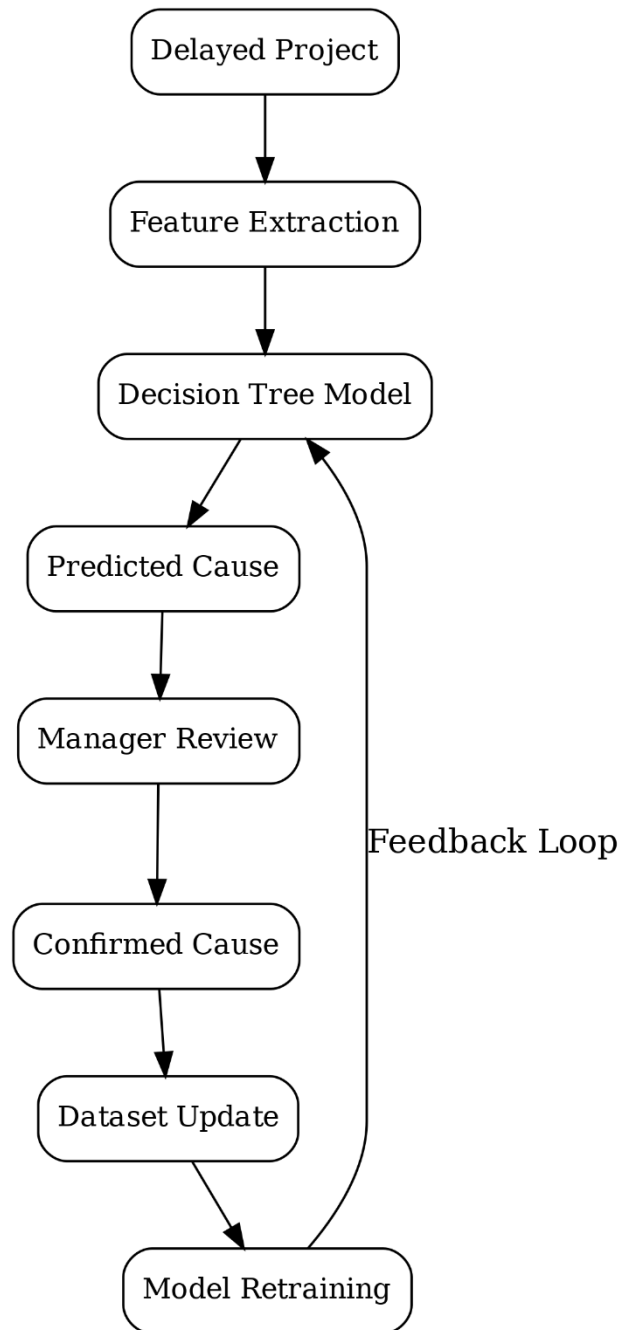


Fig. 2. AI Delay Cause Analysis Workflow (This figure outlines the end-to-end process: Project Data → Dataset Aggregation → AI Dataset → Decision Tree Model → Predicted Delay Cause → Manager Validation → Confirmed Delay Cause).

The integration of Artificial Intelligence into enterprise workflows necessitates a rigorous Explainable AI (XAI) framework. In the context of project management, models that maximize predictive accuracy at the expense of transparency are often impractical, as stakeholders cannot validate or confidently act upon opaque conclusions. Consequently, the proposed system fundamentally prioritizes explainability over raw algorithmic complexity. The Decision Tree algorithm serves as the cornerstone of this approach, providing a human-readable structure that delineates the exact operational metrics leading to a specific delay cause. By explicitly mapping features—such as workflow approval latencies or personnel volatility—to historical outcomes, the system translates abstract data into actionable business intelligence.

Crucially, this architecture embeds a strict Human-in-the-loop (HITL) methodology. The AI is positioned exclusively as a decision support mechanism. When the model infers a potential delay cause, it presents this hypothesis alongside the supporting evidence to the project manager. This evaluative paradigm ensures that human intuition, contextual awareness, and tacit knowledge—elements often invisible to the algorithm—remain central to the final determination. The project manager bears the ultimate responsibility for validating, modifying, or rejecting the AI-generated hypothesis. This validation step is not merely an administrative formality; it acts as a continuous data-cleansing loop. Only manager-confirmed delay causes are synchronized back into the training dataset. This governance rule guarantees that the machine learning pipeline is consistently nourished with verified ground-truth labels, mitigating algorithmic bias and fostering profound trust between human practitioners and the AI support system (Rayegan & Menzies, 2026; Schetinin et al., 2005; Park, 2026).

5 System Evaluation and Discussion

Initial observations regarding the architectural design of the AI-Integrated Project Management System suggest potential improvements over traditional, siloed tracking methodologies. The system is designed to mitigate the data fragmentation commonly observed in cross-organizational and multi-team environments by centralizing task management, budget tracking, and progress reporting into a unified platform. By enforcing strict state transitions and approval workflows, the architecture facilitates the collection of highly accurate and temporally precise operational data, ranging from personnel volatility to average approval latencies. This structured data foundation is essential for supporting subsequent analytical and AI-driven processes.

In evaluating the analytical framework, the application of the Decision Tree model for delay cause analysis appears to provide a viable mechanism for extracting interpretable insights. Utilizing the 22 engineered features, the model architecture is designed to identify probable bottlenecks within delayed projects. Preliminary structural analyses indicate that features such as personnel changes and approval latency serve as critical splitting criteria for categorizing causes like Lack of Personnel and Approval/Process Delay. The inherent transparency of the Decision Tree's IF-THEN pathways enables project managers to conceptually trace algorithmic outputs back to specific operational metrics. This approach has the potential to streamline the manual

auditing of extensive event logs, sup-orting previous literature that emphasizes the utility of interpretable classifica-tion models in practical software engineering and business contexts (Johannes & Alamsyah, 2021; Tantithamthavorn et al., 2020).

Table 5. Performance of the Decision Tree Delay Cause Analysis Model.

Metric	Value
Accuracy	86.96%
Macro Precision	87.66%
Macro Recall	86.94%
Macro F1-score	87.03%
Weighted F1-score	87.10%
Training Samples	368
Test Samples	92
Number of Classes	10

The evaluation results indicate that the Decision Tree model achieved stable classification performance across the defined delay-cause categories. The obtained Accuracy and F1-score values suggest that the model can provide meaningful decision-support information for project managers. However, these results should be interpreted as internal validation outcomes because the dataset was derived from the operational environment of the proposed system.

A distinguishing characteristic of this proposed solution is its reliance on a Human-in-the-loop (HITL) architecture. The framework does not autonomously execute corrective actions, such as altering schedules or reassigning resources, based on algorithmic outputs. Rather, it preserves the business authority of the project manager, deploying the AI component strictly as a diagnostic support tool. This configuration is intended to mitigate risks associated with automation bias and to ensure that complex, unquantifiable contextual factors are integrated into the final root cause analysis. Additionally, the mandatory confirmation step functions as an ongoing data-validation mechanism. Appending only human-validated delay causes to the historical dataset theoretically protects the model from the compounding errors of label noise and spurious correlations, issues that frequently compromise the reliability of long-term machine learning deployments (Rayegan & Menzies, 2026).

Despite its theoretical advantages, the current system design exhibits notable limitations. The predictive reliability and generalization capacity of the Decision Tree model are fundamentally constrained by the availability and quality of historical data. During early deployment phases, or within environments possessing a low volume of completed projects, the system may fail to satisfy the minimum training gating conditions, thereby temporarily limiting the applicability of automated AI insights. Furthermore, while the 22 tabular features encapsulate a broad snapshot of project friction, they are primarily aggregated metadata. The current tabular machine learning approach does not natively capture the complex topological dependencies between interrelated tasks, nor does it fully map the sequential, temporal progression of workflow events. Finally, while literature suggests that ensemble kernel methods can enhance robustness in high-

dimensional classification tasks (Feng & Baumgartner, 2020), the current framework deliberately prioritizes the interpretability of a single decision tree over the potential theoretical accuracy improvements offered by more opaque ensemble techniques.

Future extensions of this research present several avenues for methodological and functional enhancement. The comprehensive event logs generated by the system's operational layer provide a suitable substrate for the application of advanced Process Mining techniques. Such methods could potentially be leveraged to map process deviations and visualize localized workflow bottlenecks dynamically (Dorado et al., 2025). Moreover, a transition from static, post-mortem root cause analysis to dynamic, mid-execution forecasting may offer significant benefits for project risk management. The integration of sequence-aware or topological algorithms could enable the system to interpret the spatial dependencies of task networks and the temporal evolution of delays, theoretically facilitating proactive predictions of schedule and cost variances during active execution (Mirjalili et al., 2025; Kula et al., 2023). Nevertheless, any future incorporation of complex, non-linear algorithms must be carefully balanced with robust explainability frameworks to preserve the analytical transparency and user trust established by the current decision tree methodology.

6 Conclusion

Project delays remain a major challenge in modern organizations, often leading to cost overruns, inefficient resource utilization, and reduced project quality. While many existing studies focus primarily on predicting schedule deviations, understanding the underlying causes of delays is equally important for supporting effective project management and corrective decision-making. Consequently, organizations require not only predictive capabilities but also analytical mechanisms that help managers identify and understand the factors contributing to project delays. This study proposed an AI-Integrated Project Management System that combines operational project management with AI-assisted delay cause analysis. By centralizing project execution data, workflow activities, resource information, progress records, and cost-related data, the proposed system establishes a structured foundation for analytical decision support. An interpretable Decision Tree model was employed to identify potential delay causes from historical project patterns. Unlike black-box approaches, the selected model provides transparent decision rules that allow project managers to understand the reasoning behind AI-generated results.

A key contribution of this research is the integration of Explainable AI within a Human-in-the-Loop framework. Rather than replacing managerial judgment, the AI component functions as a decision-support mechanism that generates evidence-based hypotheses regarding possible delay causes. Project managers remain responsible for reviewing, validating, and confirming these analytical outcomes before they are incorporated into future datasets. This approach promotes transparency, accountability, and trust while ensuring that organizational knowledge remains under human control. The findings demonstrate that integrating explainable AI into a practical project management environment is both feasible and beneficial for supporting delay cause analysis.

Nevertheless, the effectiveness of the proposed approach depends on the availability and quality of historical project data, and the current feature representation may not fully capture complex workflow dependencies. Future work will focus on expanding the project dataset, incorporating workflow-oriented analytical techniques such as process mining, and investigating more advanced yet interpretable models to further enhance delay cause analysis and managerial decision-support capabilities.

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